

Exam Date & Time: 28-Dec-2024 (02:30 PM - 05:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

Manipal Institute of Technology, Bengaluru
BTech VII Semester - Make up Examination - December -2024

Advanced Cryptography [CSE-4059]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Calculate the order of the group G formed by the integers modulo 5 under addition, Z_5 . (5)
 - A)
 - B) Prove that the set $G = \{1, 2, 3, 4, 5, 6\}$ is finite abelian group of order 6, with respect to multiplication modulo 7. (5)
- 2) Use Euler's theorem to calculate $7^{100} \bmod 40$. (5)
 - A)
 - B)
 - i. Use Fermat's Little Theorem to find $3^{100} \bmod 7$.
 - ii. Compute the Legendre symbol $\frac{3}{7}$. (5)
- 3) In a RSA cryptosystem, a participant uses first odd and fifth prime numbers to generate his private and public keys. If the private key is 7, then how will the text COMPUTER be encrypted & decrypted. (5)
 - A)
 - B) Given the elliptic curve $E : y^2 = x^3 + 2x + 3$ over F_7 and the point $P=(3,6)$, compute $2P$ (Point doubling). (5)
- 4) Given an ElGamal digital signature system with prime $p=19$, generator $g=10$, and Alice's private key $x=8$, compute the public key. Then, show the process of Alice signing a message $m=6$ using a random value $K=5$. Explain how Bob verifies the signature. (5)
 - A)
 - B) Suppose you are using a 64-bit hash function for your digital signature scheme. Using the birthday paradox, estimate the number of hashes you need to generate to have a 50% chance of finding two messages with the same hash. Explain why this poses a security concern. (5)
- 5) Alice receives Bob's X.509 certificate, which contains his public key and is signed by a (5)

Certification Authority (CA). If the CA's public key is $e=3$ and $n=187$, and the signature on Bob's certificate is 110, show how Alice can verify the signature. Assume Alice retrieves the hash of Bob's certificate as 12.

A)

B)

Scenario: An e-commerce platform has implemented a public key infrastructure (PKI) to secure transactions between buyers and sellers. The platform uses digital certificates issued by a trusted certificate authority (CA) to authenticate users and encrypt transactions.

- i) Explain the key components of the PKI used by the e-commerce platform. How do these components work together to ensure secure communication between users?
- ii) Imagine a situation where a seller's private key is compromised. Discuss the role of the certificate revocation list (CRL) and how the e-commerce platform can notify users about the revocation of the seller's digital certificate. What steps should users take to ensure their transactions remain secure?

(5)

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